

Austin Billings

SA 1(HSYD301-1)

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Question (Option 1)

System development lifecycle to improve waste management systems

Nexus Innovate Tech plans to develop a Smart Recycling and Waste Management System to improve waste disposal and recycling in urban communities. The current system is inefficient because waste collection is not properly monitored, collection routes are poorly managed, and communities are not motivated to recycle. This led to appointing a System Development Specialist. SDLC ensures that solutions are designed efficiently

1. Investigation phase

Statement of the Problem

-The waste management company still uses traditional processes that are slow and ineffective. Many areas are left with uncollected waste; resources are not used efficiently, and there is no centralized system to monitor operations. Residents are also discouraged from recycling because there are no rewards or incentives.

Why the System is Being Designed

The system is being designed to:

- Reduce landfill waste through better recycling.
- Improve waste collection using AI and IoT technologies.
- Encourage communities to recycle through reward points.
- Provide a centralized system for monitoring waste management activities.

Who Will Use the System

- Residents will use mobile apps to scan recyclable items and earn rewards.
- Waste collection employees will use the system for collection schedules and route updates.
- Municipal managers will monitor waste levels and operational performance through dashboards.

Capabilities of the System

The system will provide:

- AI-powered waste classification.
- Real-time smart bin monitoring using IoT sensors.
- Route optimization for waste trucks.
- Reward points and gamification features.
- Community dashboards to track recycling progress.

Feasibility Study;

1.Economic Feasibility

- The system will reduce operational costs by improving route planning and reducing unnecessary waste collection trips.

2.Technical Feasibility

- The solution is technically possible because modern AI, IoT, cloud computing, and mobile technologies can support the system requirements.

3.Operational Feasibility

- The system will improve communication, monitoring, and waste collection efficiency for both employees and residents.

4.Scheduling Feasibility

- The project can be completed within the planned schedule using Agile development methods.

5.Legal Feasibility

The system supports environmental regulations and improves accountability in waste management.

Proposed Solution

The Waste Management Company should implement a cloud-based Smart Recycling and Waste Management System that uses AI and IoT technologies. Residents can request waste services, scan recyclable items, and earn reward points, while municipalities can monitor waste levels and optimize collection routes in real time.

Project Planning

Phase Duration

Requirements Gathering	Week 1
System Design	Week 2 -3
Development	Week 3-5 sprints agile
Testing	Week 4
Deployment	Week 7

2. Analysis phase

The Procedure for Collecting Data;

To gather accurate system requirements for the smart waste management system, several data collection techniques will be used to ensure that the solution addresses real community and business need effectively

-Interviews

Interviews will be conducted with stakeholders such as waste management employees, municipal representatives, community members, sponsors, and technical teams. These meetings will help gather valuable insights about the current challenges in waste collection, recycling processes, resource allocation, and service delivery. By directly engaging with stakeholders, the system can be designed according to real operational requirements and practical business use cases.

-Online Meetings and Collaboration

Online communication platforms such as Microsoft Teams, WhatsApp, and GitHub will be used to support collaboration between developers, project managers, and stakeholders. These platforms will allow teams to share updates, discuss system improvements, test features from a centralized repository, and maintain continuous communication throughout the development process. GitHub will also assist with version control and collaborative software development.

-Surveys and Questionnaires

Online surveys and physical questionnaires will be distributed to community members and waste management staff to collect public opinions and feedback. These surveys will help identify areas where waste management services are lacking and where additional resources are required. The collected data can also be used to classify recycling behavior, understand community participation levels, and improve future planning.

-Observation

Observation techniques will be used to physically assess current waste management operations and processes. This involves monitoring how waste is collected, transported, classified, and managed in different areas. By observing operations directly, the project team can better estimate resource requirements, identify inefficiencies, and understand real-world challenges. Observation also supports Agile development by allowing continuous improvement based on practical findings during the project lifecycle.

-Document Analysis

Existing reports, regulations, operational records, and environmental policies will be analyzed to gather detailed information about waste management processes and legal requirements. Data analysts and information teams may also use AI-powered analytical tools to identify trends, opportunities for improvement, and areas where efficiency can be enhanced. This process helps ensure that the system aligns with environmental standards and business objectives.

Defining the System Requirements

Functional Requirements:

-AI-Powered Waste Classification

The system will use AI technologies such as Amazon Web Services SageMaker to analyze waste data and classify different types of waste materials. Images and input data collected from users or IoT devices will be processed to provide accurate waste classification. This information will assist waste management teams in improving recycling efficiency and resource allocation while maintaining transparency through evidence-based reporting.

-Gamification and Rewards System

The platform will include a gamification feature that rewards users for participating in recycling and waste management activities. Sponsors and partner organizations can provide vouchers, rewards, or incentives to encourage community involvement. Services such as AWS Lambda can trigger automated events whenever users complete recycling activities, ensuring fair and transparent point allocation.

-Real-Time IoT Monitoring and Route Optimization

Employees and field workers will use mobile applications or IoT-enabled devices to manage live waste collection requests. The system will monitor waste levels in real time and send alerts for areas requiring urgent attention. Route optimization features will help drivers and waste collection teams travel efficiently, reducing fuel costs, improving service delivery, and preventing unnecessary waste of resources.

Non-Functional Requirements:

-User-Friendly Interface

The system must provide an easy-to-use and accessible interface available through both a mobile application and a web application. This ensures that community members, employees, and stakeholders can access the system conveniently.

-Scalability

The system should be scalable enough to support growing urban populations and increasing waste management demands. Cloud computing solutions are recommended to ensure flexibility, performance, and future expansion capabilities.

-Compliance and Security

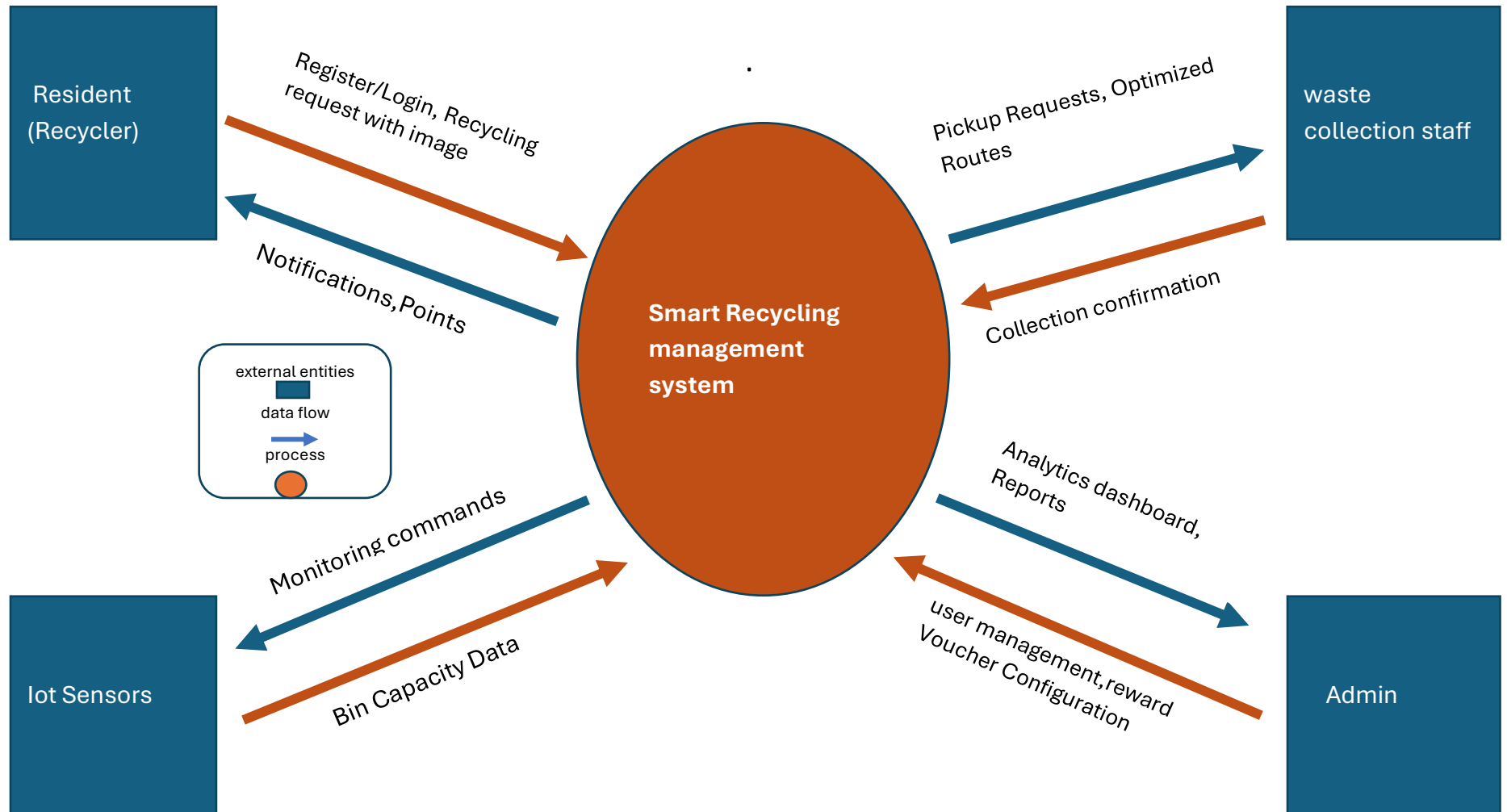
The platform must comply with environmental regulations and data protection laws such as GDPR. Dashboards and reporting systems should provide transparent evidence, including images and analytics, to demonstrate how resources and funding are being used effectively within communities and government waste management initiatives.

Prioritization of requirements

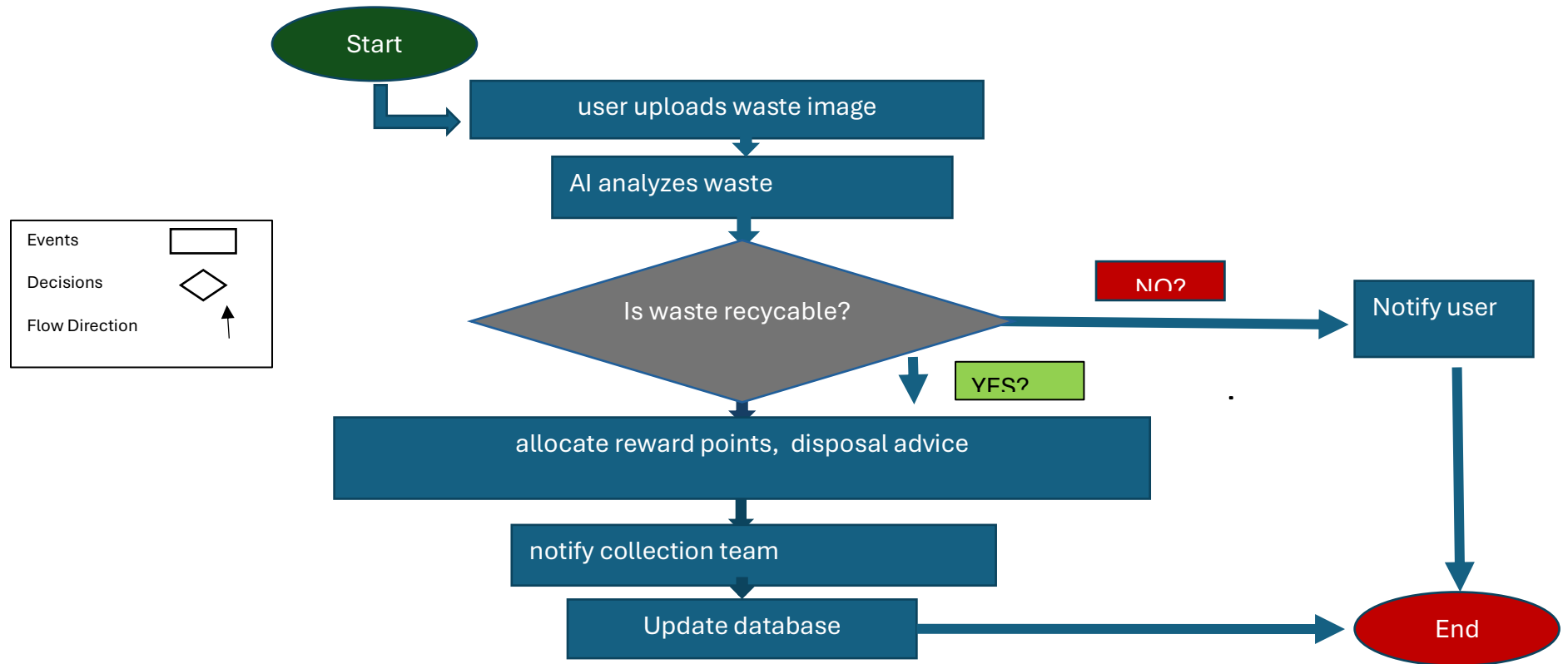
High priority	Medium priority	Low priority
--Backend infrastructure for AI image classification and monitoring systems. --Main waste management dashboard and deployment system. --IoT integration for trucks and smart waste monitoring sensors. --Real-time analysis and efficient allocation of resources.	--Frontend user interface design. --Real-time route suggestions and map directions. --Notification and trigger-based alert systems.	--Gamification features such as reward tokens, weekly leaderboards, and gift incentives. --Community achievement, celebrations, and engagement enhancements.

3. Design phase (Waste Management System)

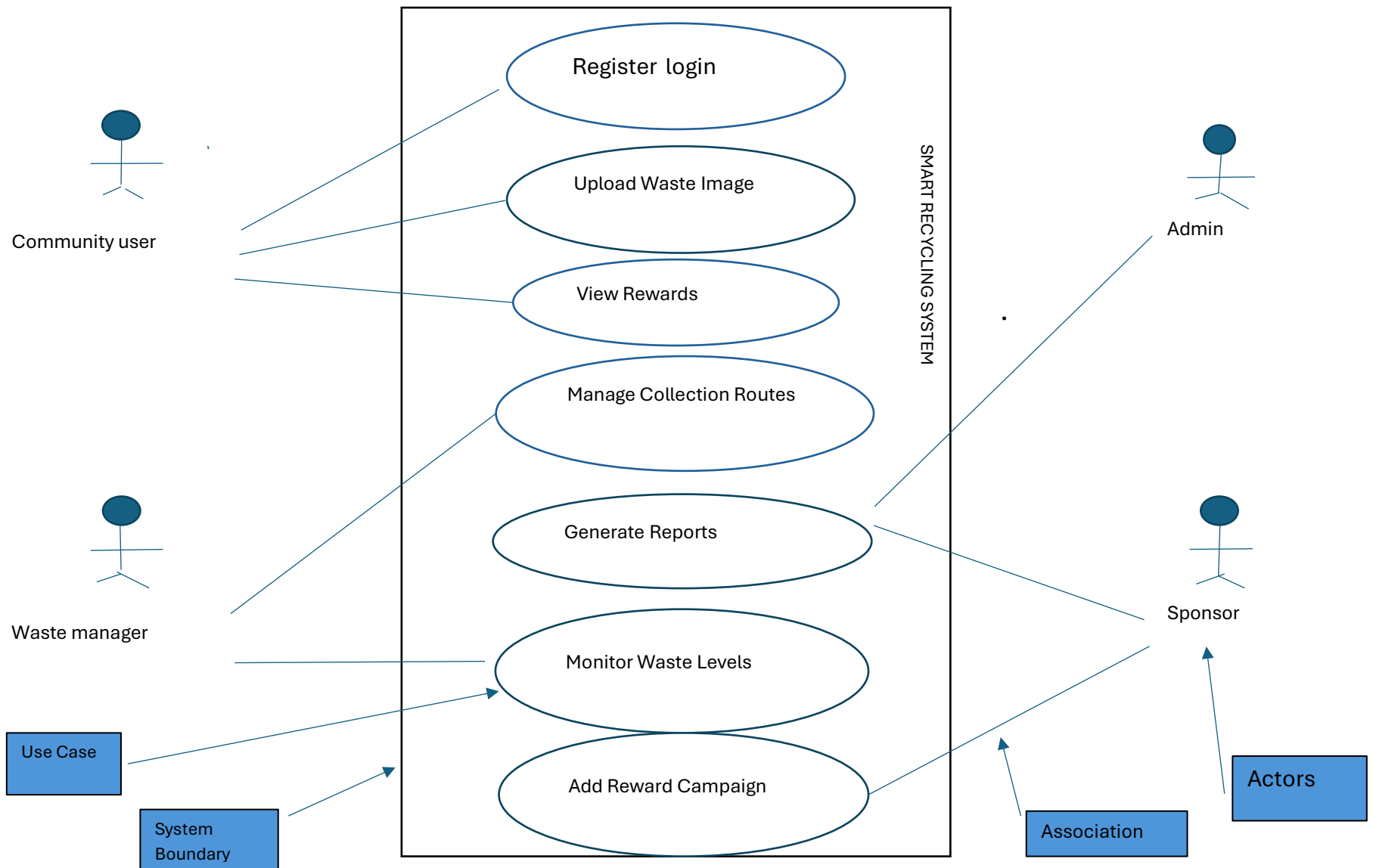
A) Context diagram



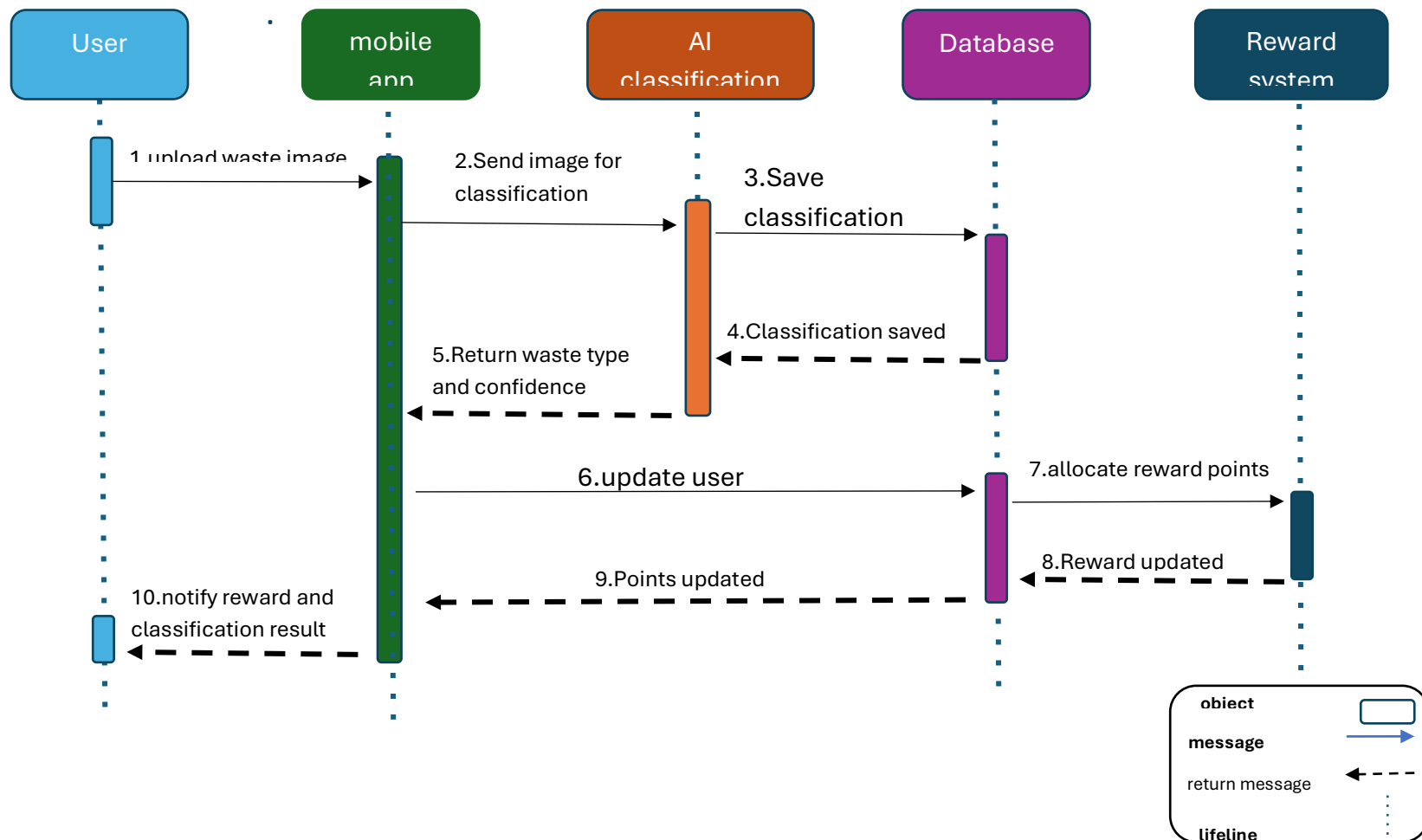
B) Flow Chart Diagram



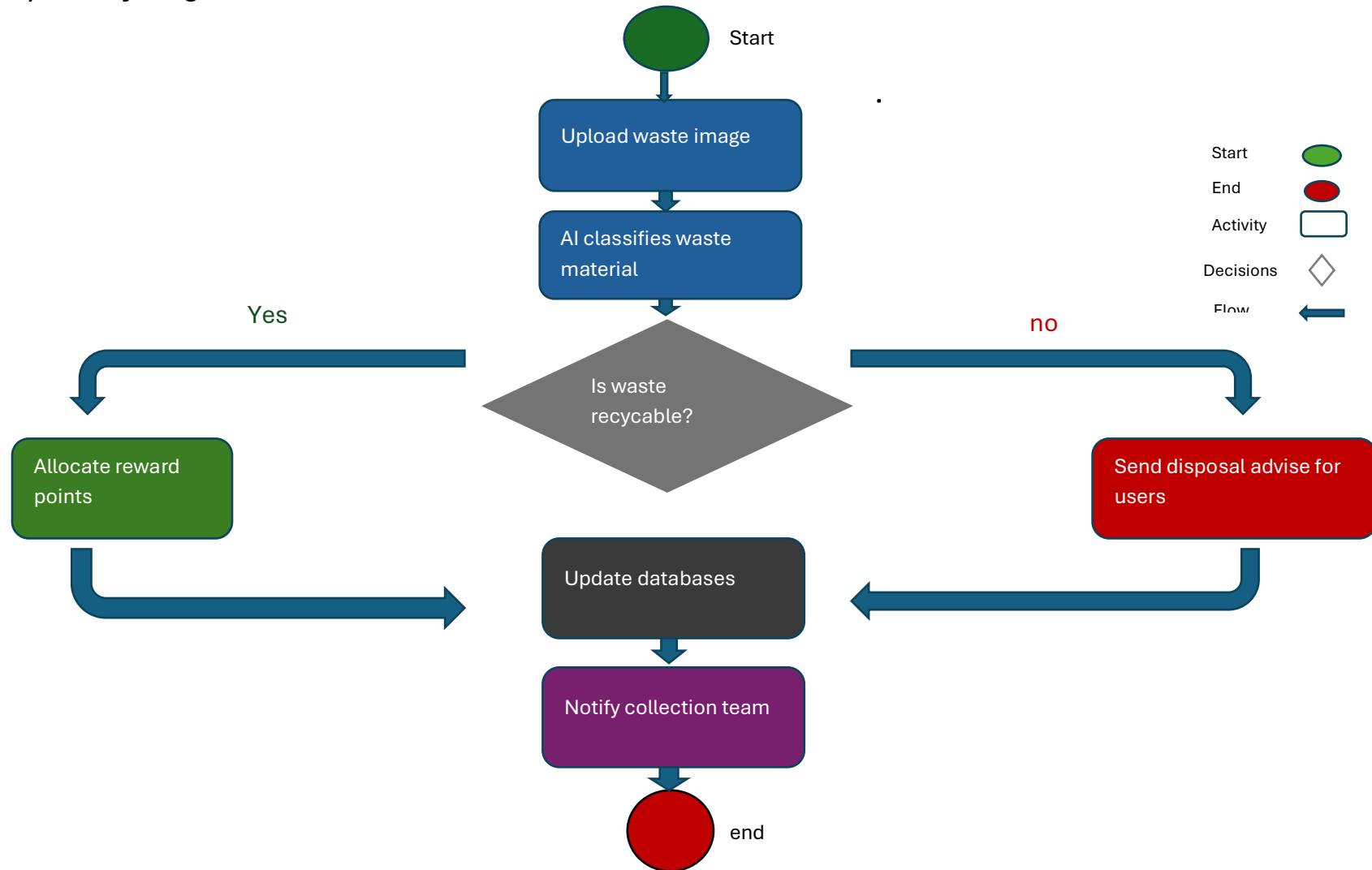
C) Use Case diagram



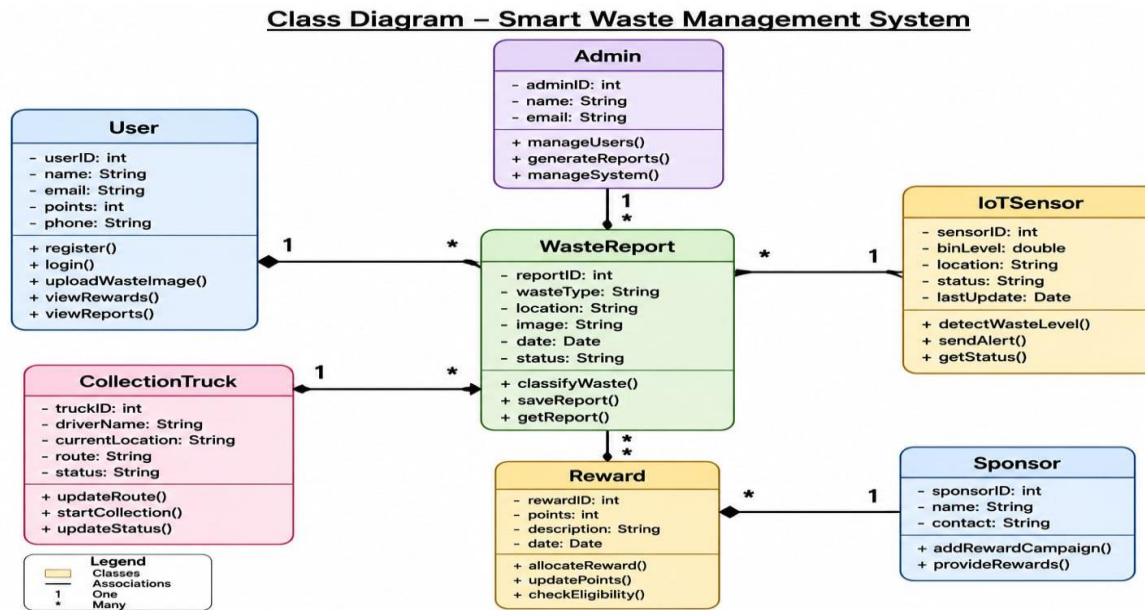
D) Sequence diagram



E) Activity Diagram



F) Class diagram



4.Build / Implementation Phase

Recommended IT Technologies:

The Smart Waste Management System will be developed using modern cloud and mobile technologies to ensure scalability, reliability, and efficient system performance.

Frontend Development

- Android Studio for mobile application development.
- HTML, CSS, and JavaScript for the web application interface.

Backend Development

- Java or Node.js for server-side processing.
- REST APIs for communication between system components.

Database

- MySQL for storing user information, waste reports, rewards, and collection records.

Artificial Intelligence

- Amazon SageMaker for waste image classification and analysis.

Cloud Services

- AWS Lambda for automated processing and event-driven functions.
- Amazon S3 for storing uploaded waste images and reports.

IoT Integration

- Smart sensors installed in waste bins to monitor waste levels and send real-time data.

Version Control and Collaboration

- GitHub for source code management and team collaboration.

Why These Technologies Are Appropriate

- Cloud services provide scalability for growing communities.
- AI improves waste classification accuracy.
- IoT sensors enable real-time monitoring.
- Mobile applications improve accessibility for users and employees.
- MySQL provides reliable data storage and management.
- GitHub supports collaborative development and version control.

User Interface Design

Selected User Interface Design and Design Consideration

The Smart Waste Management System will use a responsive and user-friendly interface that can be accessed through both mobile and web applications. The interface will provide simple navigation, allowing users, employees, administrators, and sponsors to perform their tasks efficiently.

Design Considerations

Simplicity and Ease of Use

- Clear menus and navigation options.
- Simple dashboard displaying waste reports, rewards, and collection schedules.

Accessibility

- Mobile-friendly design for community users and field workers.
- Easy-to-read fonts and icons.

Performance

- Fast loading screens and real-time updates.
- Efficient display of waste monitoring information.

Security

- Secure login and authentication.
- Protection of user data and waste management records.

External Input Metrics

The system receives data from various external sources:

1. Waste Image Uploads

- Users upload images of waste materials for AI classification.

2. IoT Sensor Data

- Smart bins send waste-level information to the system.

3. GPS Location Data

- Collection trucks provide location information for route optimization.

4. User Reports

- Community members submit reports regarding overflowing bins or illegal dumping.

External Output Metrics

The system generates outputs that support decision-making and service delivery:

1. Waste Classification Results

- AI provides recyclable or non-recyclable waste classifications.

2. Reward Notifications

- Users receive reward points and achievement updates.

3. Collection Schedules

- Employees receive collection requests and route information.

4. Management Reports

- Administrators receive analytics and performance reports for decision-making.

5. Testing Phase

Before deployment, the system will undergo testing to ensure that all components function correctly.

-Unit Testing

Individual modules such as waste classification, rewards allocation, and route optimization will be tested independently.

-Integration Testing

Testing will ensure that the mobile application, database, AI services, cloud services, and IoT devices communicate correctly.

-User Acceptance Testing

Stakeholders, employees, and community members will test the system to verify that it meets business requirements and user expectations.

6. Deployment Phase

Once testing is complete, the system will be deployed to a cloud environment. Mobile applications will be distributed to users, while administrators and employees will access the system through web dashboards.

Training sessions and user guides will be provided to ensure successful adoption of the system.

7. Maintenance Phase

After deployment, continuous maintenance will be performed to ensure optimal system performance.

Maintenance activities include:

- Fixing software bugs and errors.
- Updating AI models for improved waste classification.
- Monitoring cloud and database performance.
- Enhancing security features.
- Adding new features based on user feedback.
- Updating IoT integrations as technology evolves.

Regular maintenance will ensure that the Smart Recycling and Waste Management System remains reliable, secure, and effective in supporting community recycling initiatives.